

What Is Machine Learning? — Student Reference Card

ITPU · Introduction to Machine Learning · Topic 1 · keep this one page

The one idea

Traditional: **RULES** + DATA → **ANSWERS**

you write the logic

DATA + **ANSWERS** → **RULES**

the machine learns the logic (a "model")

Use ML when... / Skip ML when...

- ✓ rules are complex/fuzzy (faces, language, fraud)
- ✓ many cheap predictions, occasional error OK
- ✓ the environment keeps changing
- ✓ you have data that captures the pattern
- the rule is exact & known → just program it
- errors are catastrophic & must be explainable
- you have almost no data
- pure computation (sorting, converting units)

Core vocabulary (house-price example)

Term	Meaning	Example
Dataset	collection of examples	10,000 sold houses
Instance	one example / row	a single house
Feature	an input column	area, rooms, district
Target / label	the answer to predict	the sale price
Model	learned features→target rule	the price predictor
Prediction	model applied to new input	price of a new listing

The three families

Supervised"learn from an answer key" — labelled data.

Jobs: classification, regression.

spam · price prediction

Unsupervised"find patterns, no key" — features only.

Jobs: clustering, dim. reduction.

segments · anomalies

Reinforcement"learn from rewards, by doing" — agent + reward.

game agents · robots

out of scope

ML vs AI

All ML is AI; not all AI is ML. AI $\supset$ Machine Learning $\supset$ Deep Learning. A hand-coded rule engine (thermostat, A* game bot) is AI, not ML. A spam filter trained on labelled emails is ML. A calculator is neither.

Self-check: classify each — (1) predict house price (2) group songs with no labels (3) convert °C→°F (4) robot learns to walk.

Answers: 1 supervised · 2 unsupervised · 3 not ML · 4 reinforcement.

Practice & tools

Interactive classroom app (offline HTML): live regression demo, classify game, ML-vs-AI sorter, quiz. Try at home: [Google Teachable Machine](#), [R2D3 visual intro to ML](#), [Google ML Crash Course](#).